Nihal Chourasiya

Master of Technology in Computer Science
Indian Institute of Technology, Kanpur
nhlchrs@gmail.com
nihal-chourasiya | nihalChourasiya

Academic Qualifications

Degree	Institute/Board	CGPA/Percentage	Year
Master of Technology	Indian Institute of Technology, Kanpur	8.77	2025-Ongoing
Bachelor of Technology	Madhav Institute of Technology and Science, Gwalior	7.58	2021-2025
Senior Secondary	Central Board of Secondary Education	88%	2021
Secondary	Central Board of Secondary Education	94.6%	2019

Projects

AeroProxy - API gateway *(Jul 2026 - Aug 2026)*
Self Project
- Designed and developed a high-performance multithreaded **API Gateway in C++** (Boost.Asio/Beast, OpenSSL).
- Engineered core microservice capabilities: **TLS termination, request routing, load balancing** and **rate-limiting**.
- Implemented active health checking and **automatic failover/recovery** of backend instances.
- Developed a custom **TCP connection pool** between Gateway and instances to **eliminate handshake** overhead.
- Maximized throughput on keep-alive paths by **mitigating** TCP reconnection costs and **Nagle-buffering latency**.

BODH (Benchmarking Open Data Platform for Healthcare AI) (M.Tech Thesis) *(Apr 2026 – Present)*
Mentor: Prof. Nisheeth Srivastava, CSE, IIT Kanpur
- Developed a ML model Benchmarking platform for **National Health Authority** using FastAPI, React and PostgreSQL.
- Built an **automated evaluation system** using **Docker** that safely runs uploaded model inside isolated containers.
- Engineered a **storage-efficient system** that deletes models post-evaluation, **retaining only a SHA-256 hash** signed by the user's private key as cryptographic **proof of submission**.

Distributed 3D Stencil Engine & Isovalue Analyzer *(Mar 2026 - Apr 2026)*
CS633 - Parallel Computing
- Built a **distributed 3D stencil computation system** in C and MPI to track isovalues **across partitioned sub-domains**.
- Optimized large-scale grid processing by implementing **3D domain decomposition** and **halo exchanges**.
- Designed **boundary-handling for distributed processes** to compute isovalues without data duplication.
- Deployed on **PARAM Rudra** via SLURM, **scaling** dynamically **across varying node configurations**.

CDCIR - CIR using DINOv2 + CLIP *(Feb 2026 - Apr 2026)*
CS776 - Deep Learning for Computer Vision
- Built an **Image Retrieval system** that retrieves target images from a gallery using a starting image and text instructions.
- Created a **custom fusion layer** integrating **CLIP**'s text understanding with **DINOv2**'s granular visual region detection.
- Achieved competitive retrieval accuracy on the FashionIQ dataset, matching the performance of complex AI methods.
- Optimized the system's **speed** by **pre-computing and caching image features**, and **deployed** an interactive, real-time web application demo using **Gradio**.

IITK-Ride-Connect *May 2026*
Self Project
- Built a **driver discovery** web app to connect campus riders with nearby auto-rickshaws using Flask and MongoDB.
- Engineered a creative **SMS-based state management system** via SMSgateway, allowing drivers to update their availability **without requiring smartphones or mobile data**.

Technical Skills

- **Languages:** C/C++, Python, HTML, JavaScript • **Databases:** Redis, PostgreSQL, MongoDB, MySQL
- **Tools and Frameworks:** React.js, Node.js, RESTful APIs(FastAPI), MPI(MPICH), Git, Docker, Linux, LaTeX

Achievements

Qualified GATE CS 2024 and 2025.
Solved over **400 problems** and achieved a **LeetCode** rating of 1600+, placing in the top 21% of over 12M+ global users.

Positions of Responsibility

Teaching Assistant CS345 Algorithms II *Jul 2026 - Present*
Teaching Assistant ESC111/2 Fundamentals of Computing *Jul 2025 - May 2026*

Relevant Coursework

Advanced Data Structures
Cryptography Protocols
Parallel Computing*
Design & Analysis of Algorithms*
Deep Learning for Computer Vision
Introduction to Machine Learning

*Secured A grade